

# Panthaweeekan Somngam

Software Engineer

Bangkok, Thailand (GMT+7) | Open to Remote / Hybrid

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EN (Intermediate) | TH (Native)

## PROFESSIONAL EXPERIENCE

### Associate Software Engineer

May 2025 – Present

SCB TechX, Bangkok, Thailand (Hybrid)

- Developed a Go microservice handling token validation, authentication, and request routing for SCB Easy's 10M+ user platform, integrated behind Kong as part of a phased migration away from a legacy on-premise gateway.
- Designed the plugin system, routing engine, caching layer, and observability infrastructure for the new gateway, with Prometheus metrics and structured logging ready for production rollout.
- Designed and developed an enterprise asset management platform with multi-stage approval workflows, role-based access control for six user roles, financial reporting dashboards, and real-time notifications, used across the organization.
- Leveraged AI-assisted development tools across code generation, unit-testing, refactoring, architecture review, and documentation to accelerate delivery on complex features.

### Full Stack Developer (Work-Study Scholarship)

2023 – 2025

Chiang Mai University, Faculty of Engineering, Chiang Mai, Thailand

- Designed and deployed two production web applications serving 500+ active users, owning the full lifecycle from stakeholder requirements through containerized deployment.
- Developed a real-time room reservation system with role-based access, admin dashboard, and mobile push notifications, adopted by the faculty for daily scheduling.
- Developed a multi-stage certificate approval workflow that digitized a paper-based process, reducing turnaround from days to hours.
- Applied hexagonal architecture across all projects, building RESTful APIs in Go with React/TypeScript frontends and PostgreSQL.

## PROJECTS

### API Gateway Microservice — SCB TechX

2025 – Present

Go, Fiber, PostgreSQL, Redis, Kong, Prometheus, Kubernetes

- Built a token service and routing migration layer in Go behind Kong to incrementally replace a legacy gateway, implementing token authentication, plugin-based request transformation, and per-service circuit breaking with retry and exponential backoff across 30+ microservices.
- Developed a declarative config sync system based on the Kong deck pattern, using YAML route definitions with diff, rollback, and response caching — enabling GitOps-style config management and safe phased rollouts.
- Structured the service using Hexagonal Architecture with strict layer separation, dependency injection, and interface-driven design — keeping core business logic fully testable and decoupled from Kong, Redis, and all infrastructure dependencies.

### Inventory Asset Management System — SCB TechX

2025

Ruby on Rails, React, TypeScript, PostgreSQL, Azure AD, Docker

- Built a database-driven workflow engine to handle complex request lifecycles across four parallel workflows (request, spare, repair, return), enforcing approval chains per role and step.
- Implemented role-based access control for six user roles with Azure AD single sign-on.
- Built reporting dashboards aggregating asset requests, utilization, financial summaries, and processing efficiency, with Excel and CSV export.
- Developed the frontend with reusable components, type-safe API clients generated from the backend specification, and full offline-capable PWA support with push notifications.

## SD-Booking — Room Reservation System — Chiang Mai University

2023 – 2024

Go, React, TypeScript, PostgreSQL, PWA, Docker

- Built a real-time room booking platform serving 500+ faculty users with role-based access, booking conflict detection, and mobile push notifications, adopted for daily operations.

## FitJourney — Fitness Tracking Platform — Side Project

2026

Next.js, Supabase, Tailwind CSS, OpenAI API, Vercel

- Shipped a full-stack fitness PWA addressing a personal gap in daily tracking — consolidating workout logging, nutrition, body metrics, and AI-powered meal scanning into a single self-hosted tool.
- Designed analytics dashboards for performance trends and calorie tracking; architected as a monorepo with Supabase (auth, DB, storage) and Vercel edge deployment for fast cold starts.

## Twitch Chat TTS Bot — Side Project

2026

Bun, Microsoft Edge TTS, tmi.js, FFplay

- Engineered a real-time Twitch TTS bot that pipes synthesized audio directly from memory to playback, eliminating disk I/O and reducing per-message overhead versus file-based approaches — supporting Thai and English neural voices via Microsoft Edge TTS.
- Built a message priority queue with anti-spam filtering and automated OAuth token refresh to handle high-volume chat reliably without manual intervention.

## EDUCATION

### Bachelor of Engineering in Computer Engineering

Graduated 2025

Chiang Mai University

Second-Class Honours GPA: 3.32/4.00

Thesis: *LongPlan* — Study plan validator using graph algorithms (topological sort, cycle detection) and curriculum rule enforcement. Built in Go with React frontend.

## SKILLS

**Languages:** Go, TypeScript, JavaScript, Ruby, Python, SQL, Java, HTML/CSS, Haskell

**Frameworks:** Fiber, Gin, Ruby on Rails, React, Next.js, Node.js, Vite, Tailwind CSS, MUI, TanStack Query

**Infrastructure:** PostgreSQL, Redis, Docker, Kubernetes, AWS, Kong API Gateway, CA Layer 7, Nginx, Supabase, Vercel, Linux

**Observability:** Prometheus, Grafana, ELK Stack

**Tools:** Git, GitHub Actions, Jenkins, HashiCorp Vault

**Practices:** System Design, Data Structures & Algorithms, REST APIs, Microservices, CI/CD, TDD, SOLID, OOP, Agile/Scrum